

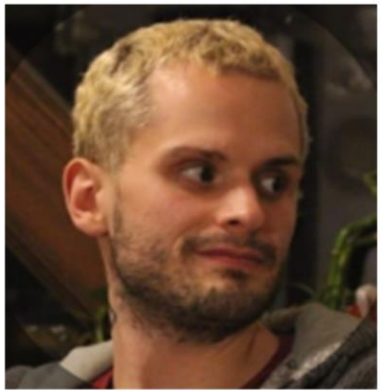


**wellcome
centre
integrative
neuroimaging**



INTRODUCTION TO OPEN SCIENCE

Why it is necessary and how we
practice open science @ WIN



Juju Fars
(They/Them/Theirs)
Postdoctoral Research
Fellow – Vision group
Legacy Open Science
Ambassador

What are the issues/threats to reproducible science?

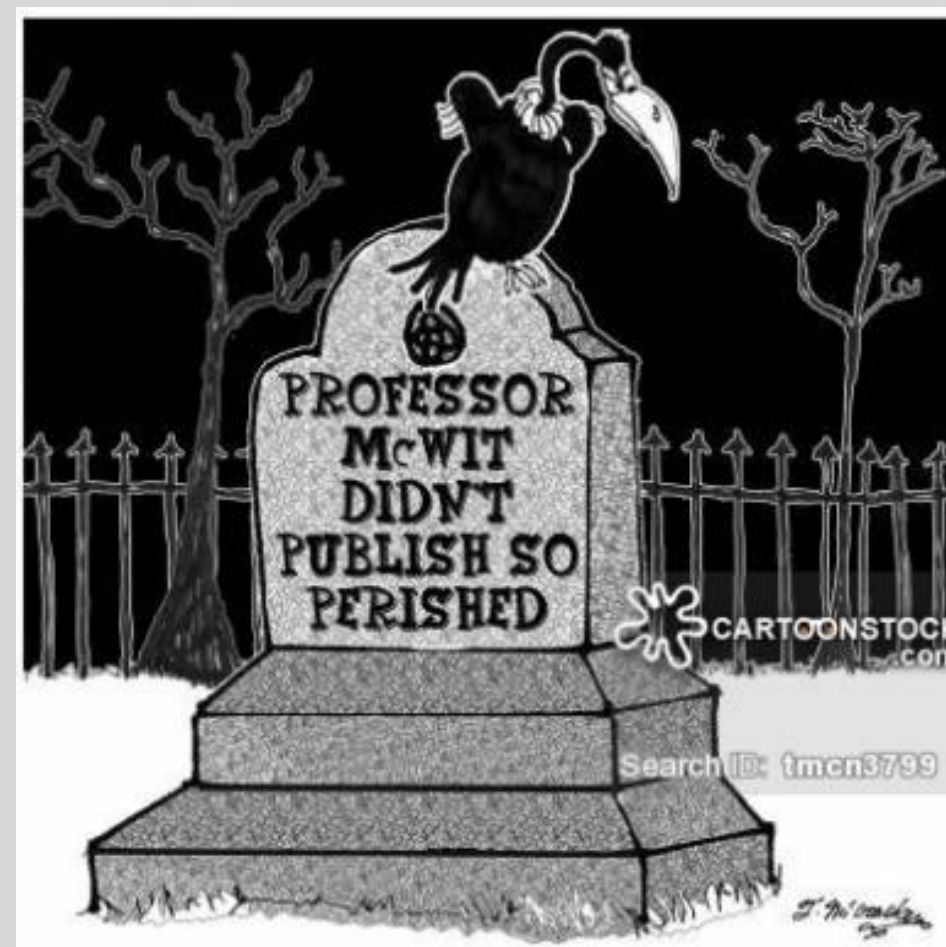
Reproducibility? Replicability?

- To follow Open Science guidelines, you need your data analysis to be REPRODUCIBLE

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Problem: Incentive Structures

- Publish regularly
- Publish “exciting” results in a high impact journal
- Make sure your work is cited
- Replications are not exciting
- Null results are not exciting
- Tell a story which supports your hypothesis



Problem: Publication bias

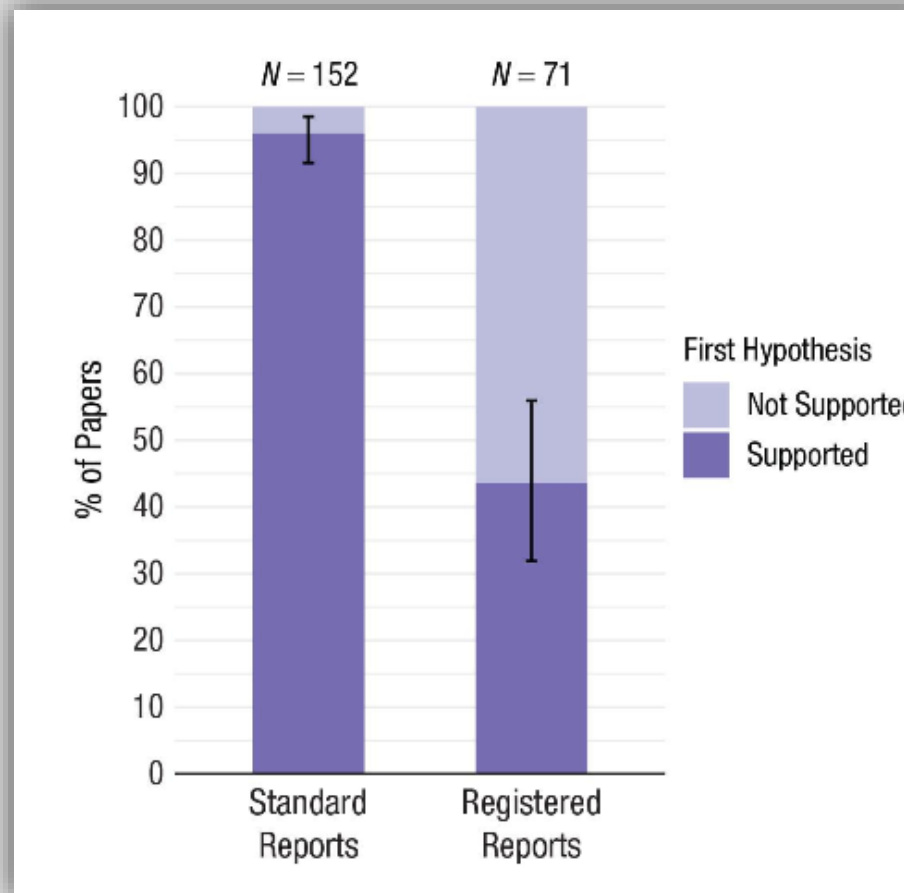
- Randomized clinical trials with positive findings (compared to negative or null):
 - Four times more likely to be published
 - Published 2-3 years earlier
 - More likely to be published in English
 - More likely to be published in higher impact factor journal.



Murad MH, Sultan S, Haffar S, et al. Methodological quality and synthesis of case series and case reports. BMJ Evidence-Based Medicine 2018;23:60-63.

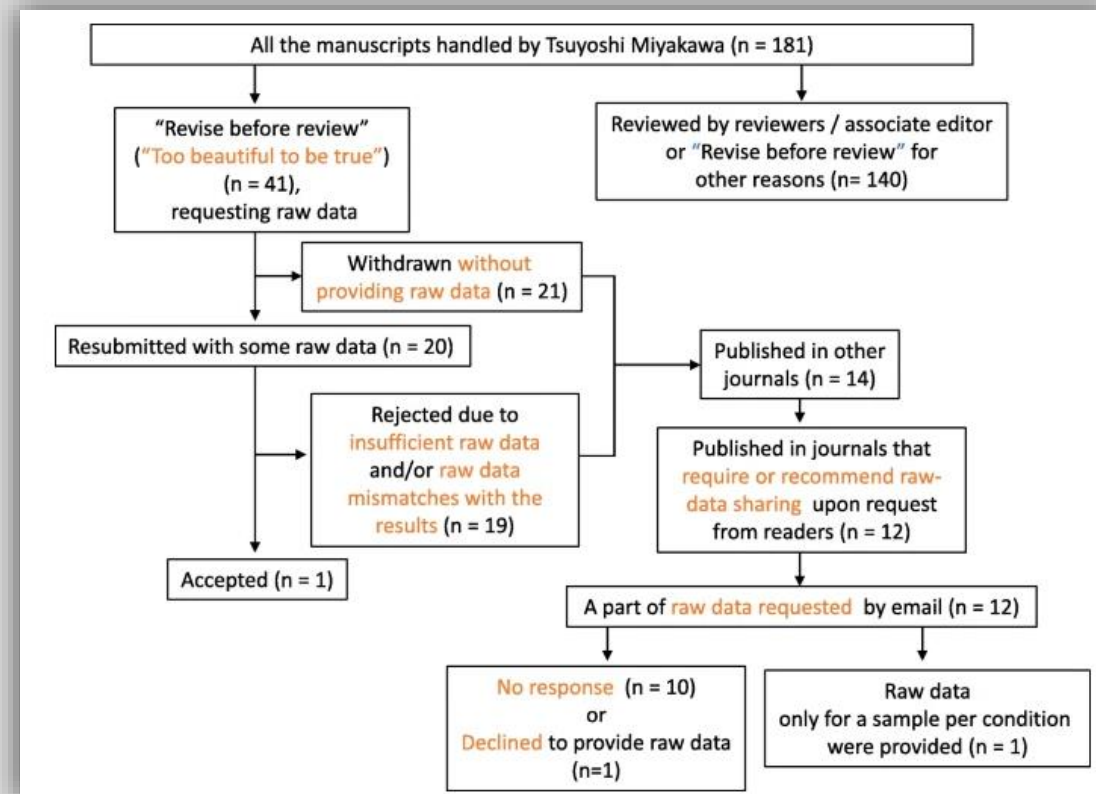
Problem: Result-based hypotheses

Regular papers are
rewritten to conform to
the results



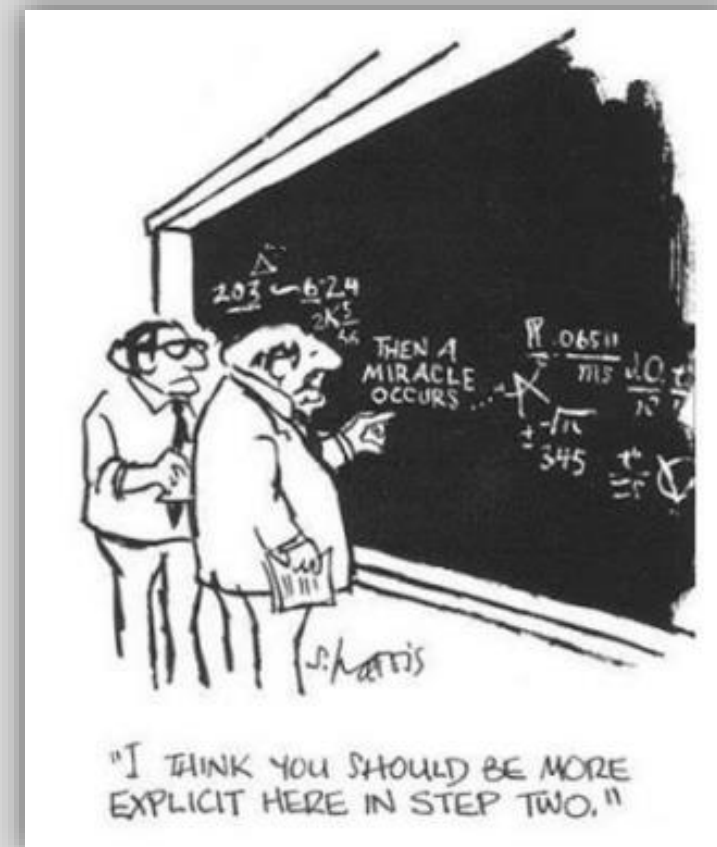
Problem: No raw data

- “Requiring raw data drives away [...] manuscripts”
- Approximately 53% of the 227 respondents from the life sciences field answered that they suspect more than two-thirds of the manuscripts that were withdrawn or did not provide sufficient raw data might have had fabricated the data.
- **Absence of raw data means the absence of science**



Problem: Poor data analysis

- “Our ability to generate data has grown dramatically, our ability to understand them has not developed at the same rate” (Peng, 2015)
- Numerous fields of research now collect an enormous amount of data but is not able to process it correctly
- Many journals do not enforce the access to power analyses

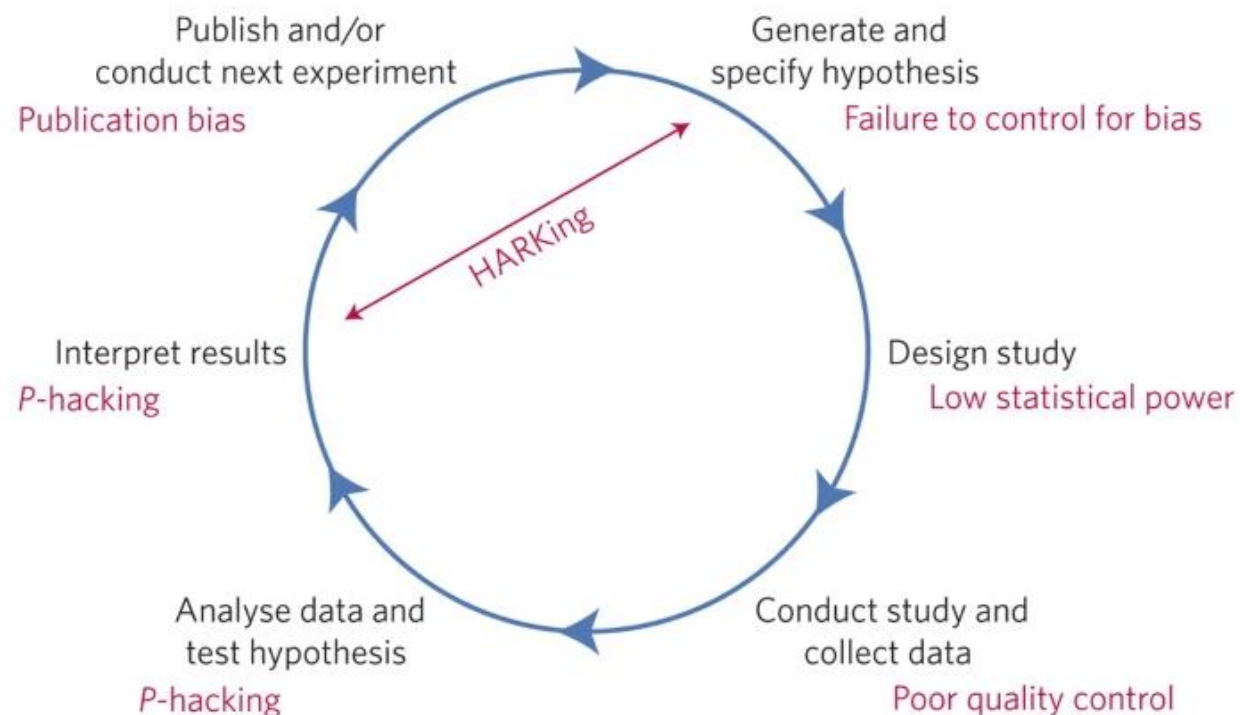


<https://www.teachthought.com/education/show-your-work/>

Problem: Fraud

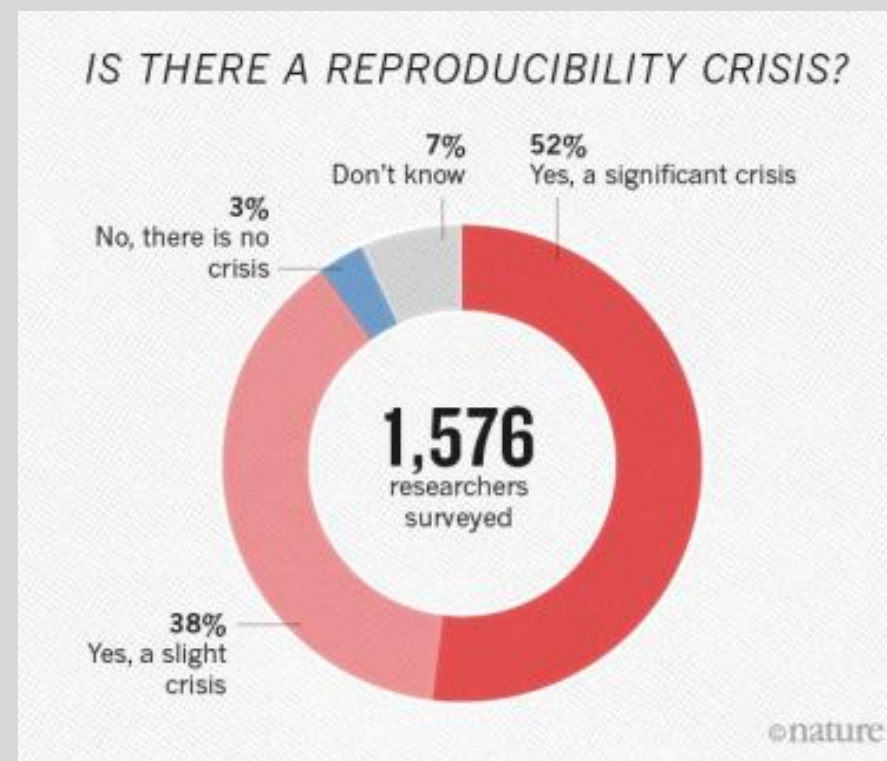
All steps in research process are subject to fraudulent practices

Figure 1: Threats to reproducible science.



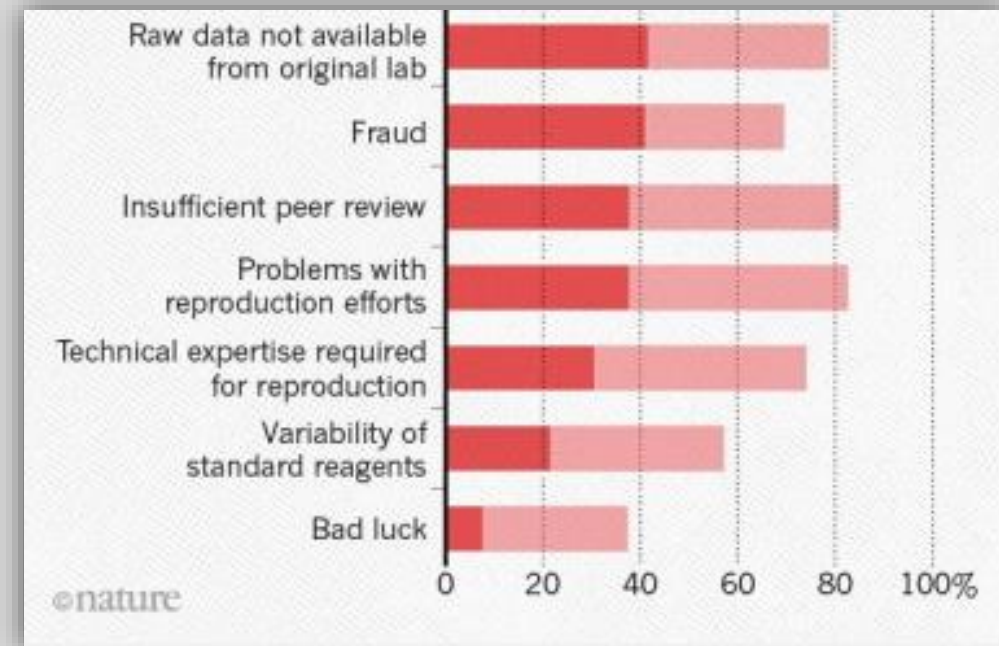
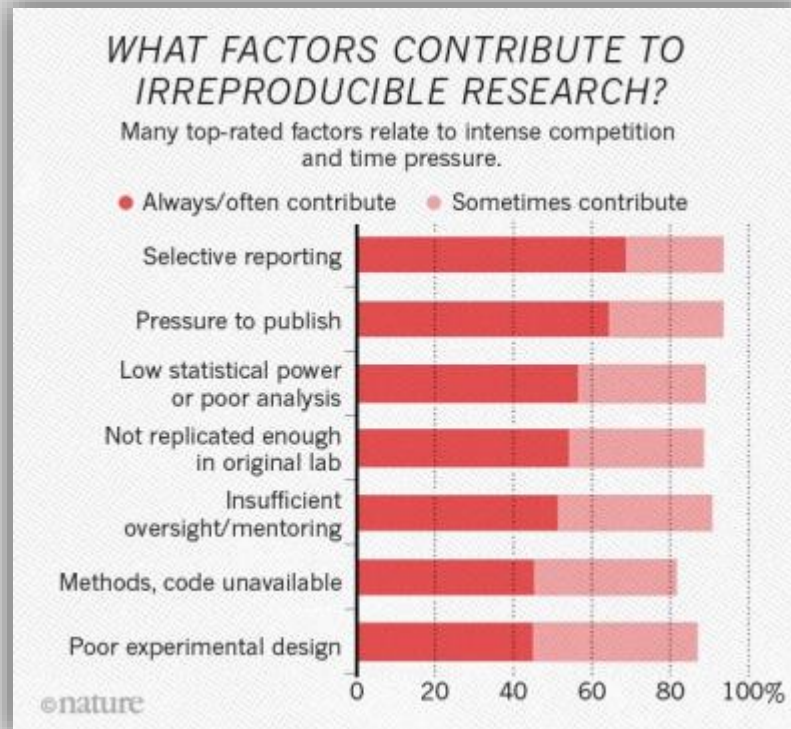
Leads to 'Reproducibility Crisis'

- 52% of those surveyed agree that there is a significant 'crisis' of reproducibility
- 85% of biomedical research efforts are wasted
- 73% said that they think that only half of the papers in their field can be trusted



Baker, M. 1,500 scientists lift the lid on reproducibility. *Nature* 533, 452–454 (2016). <https://doi.org/10.1038/533452a>
Macleod, M. R. et al. Biomedical research: increasing value, reducing waste. *Lancet* 383, 101–104 (2014).

Leads to 'Reproducibility Crisis'



Macleod, M. R. et al. Biomedical research: increasing value, reducing waste. *Lancet* 383, 101–104 (2014).

Soon to be a problem (if not already)

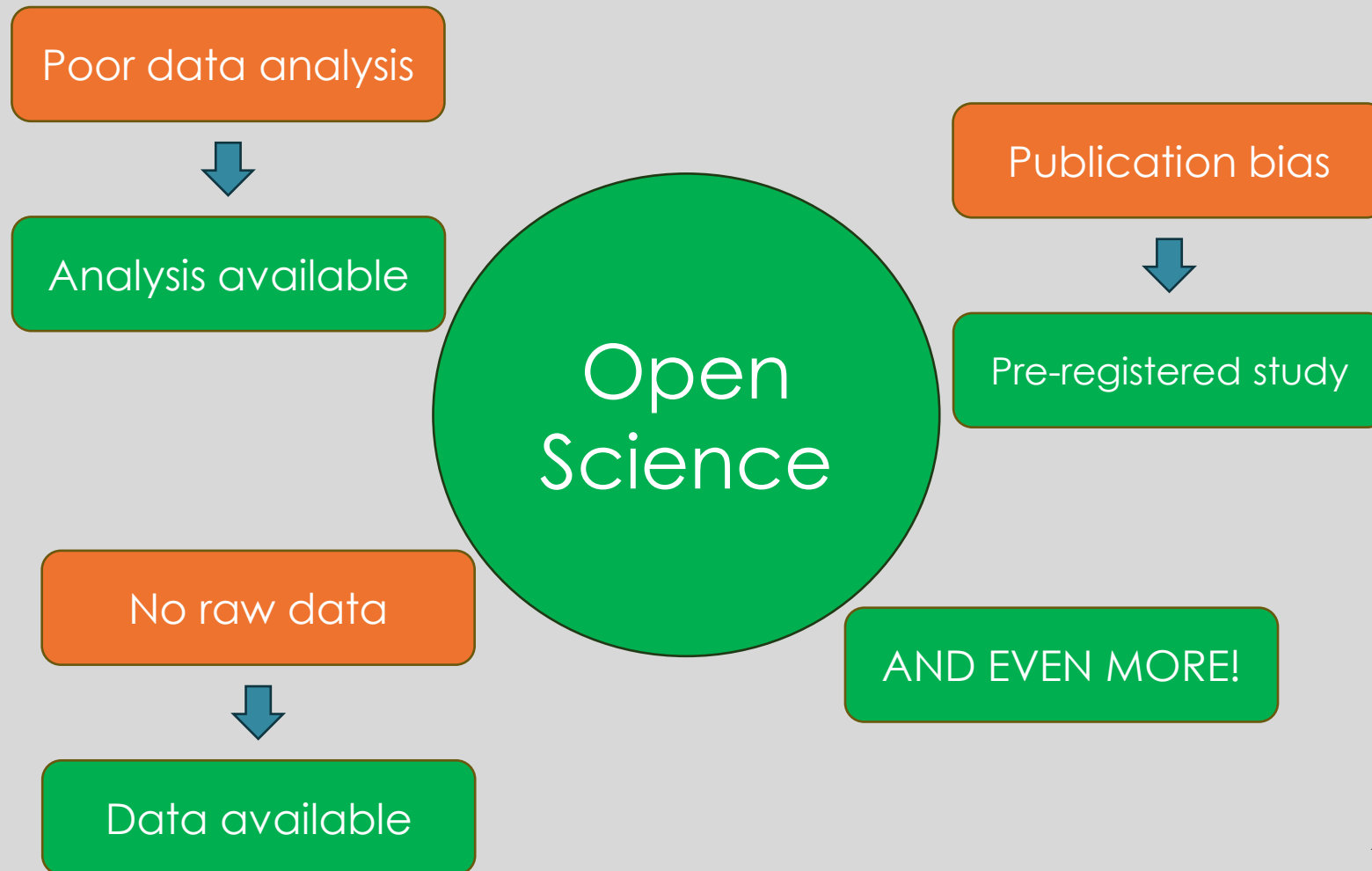
◦ Machine learning based research

- Peer reviewers do not have the time to scrutinize ML models → academia currently lacks mechanisms to root out irreproducible papers
- Over-optimism about the powers of machine-learning models could prove damaging when algorithms are applied in areas such as health and justice
- In 20 reviews in 17 research fields, results from 329 research papers could not be fully replicated because of problems in how machine learning was applied

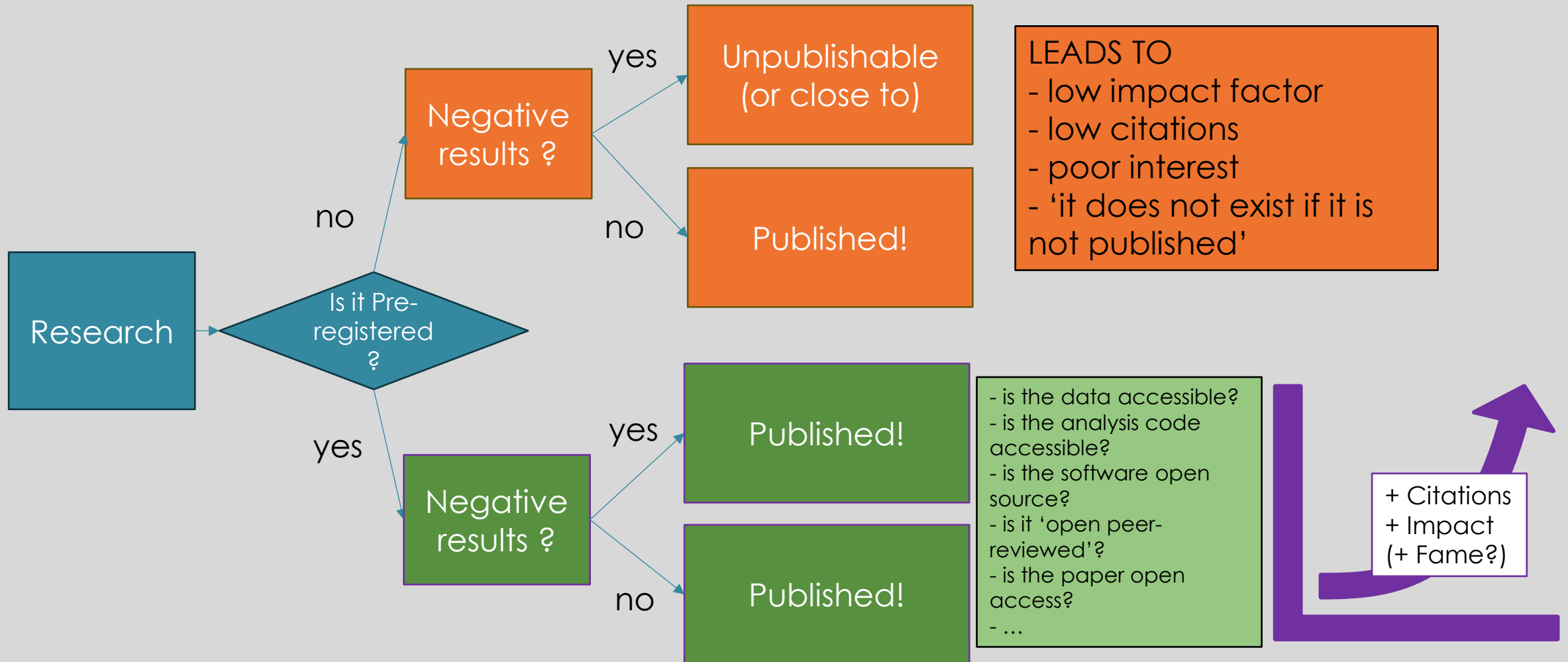
“I’m somewhat surprised that there hasn’t been a crash in the legitimacy of machine learning already. But I think it could be coming very soon.” – Momin Malik



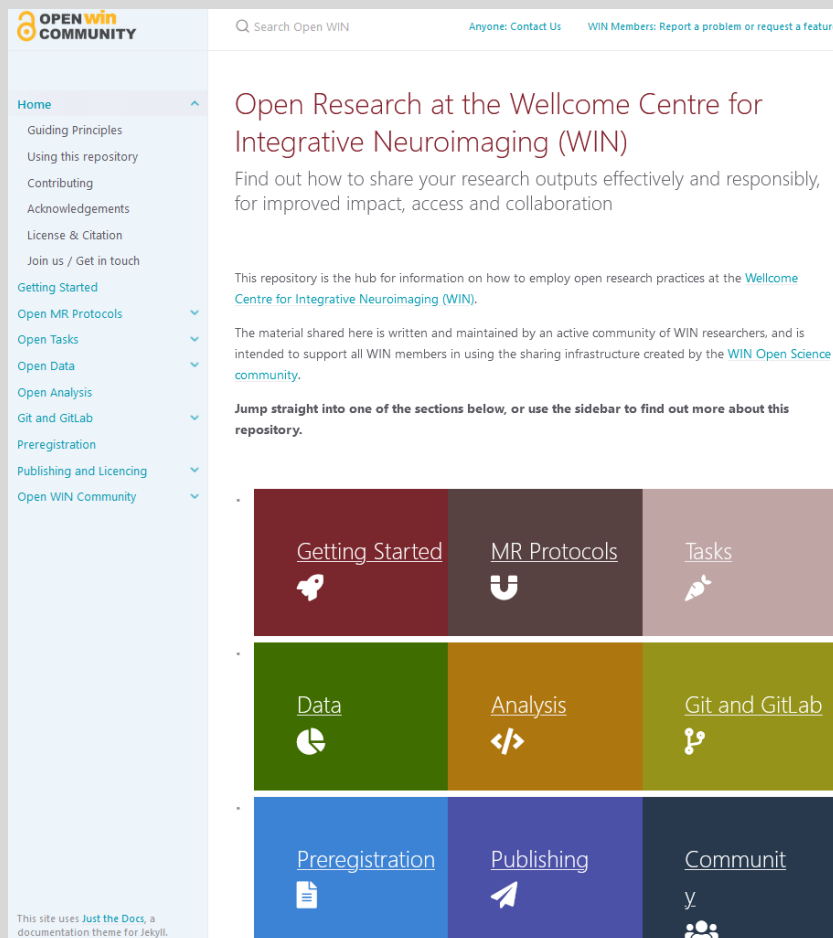
Solution: Open Science



A case study: non-OS research vs OS research



Open Win Community Pages



The screenshot shows the Open Win Community website. The header includes the logo, a search bar, and links for 'Anyone: Contact Us' and 'WIN Members: Report a problem or request a feature'. The left sidebar lists navigation options: Home, Guiding Principles, Using this repository, Contributing, Acknowledgements, License & Citation, Join us / Get in touch, Getting Started, Open MR Protocols, Open Tasks, Open Data, Open Analysis, Git and GitLab, Preregistration, Publishing and Licencing, and Open WIN Community. The main content area is titled 'Open Research at the Wellcome Centre for Integrative Neuroimaging (WIN)' and provides information on how to share research outputs. It includes a paragraph about the repository's purpose and a section titled 'Jump straight into one of the sections below, or use the sidebar to find out more about this repository.' which contains a grid of links to various sections: Getting Started, MR Protocols, Tasks, Data, Analysis, Git and GitLab, Preregistration, Publishing, and Communit.

GitLab

How to use the WIN GitLab instance for your code and documentation



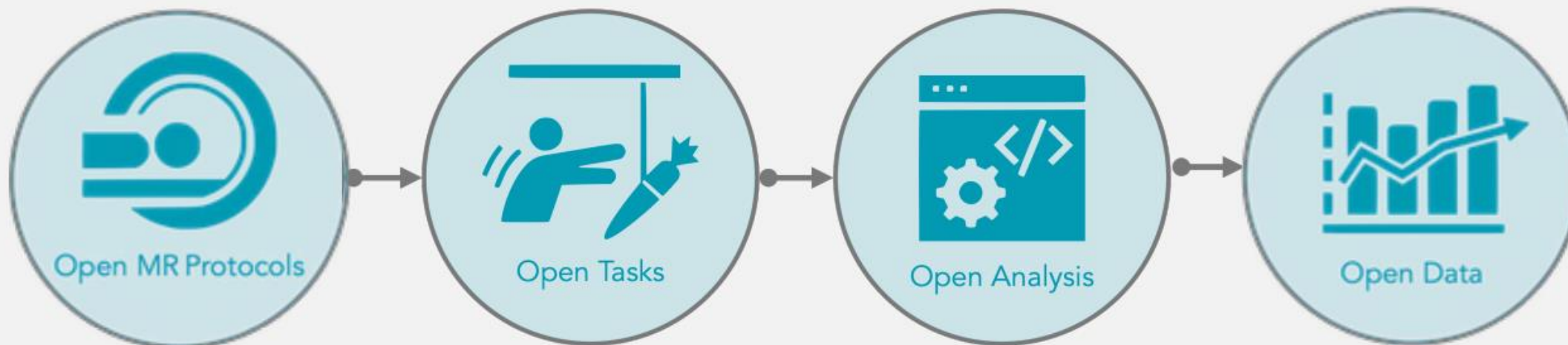
GitLab

Overview

GIT is a service which allows version control of material and syncing between local (on your computer) and remote (on a server, accessible through a web browser) versions of material. These remote and local versions are useful for backup and collaboration, where different users can access the same remote content.

WIN provide a git service using the open source [GitLab software](#). We have installed GitLab on our internal servers, so you can use it to maintain your material in a space where we have complete control over who can access it.

GitLab is best used for the management of text-based documents with some pictures (not nifti brain images!). This mainly covers code and documentation. It is also useful for project management by logging issues and milestones, and tracking collaboration.



SIEMENS
Healthineers

 **GitLab**



 **XNAT** 1.7

Open Win 2024 Ambassadors

*'incorporate
open
research
practices
into my
workflow'*



Anna Guttesen
(NDCN)



Lara Nikel
(NDCN)



Lilian Weber
(Psychiatry)



Ying-Qiu (Akina) Zheng
(NDCN)

*'advocate for
more robust
open science
practices'*

*'help promote an
open research
culture both within
and outside the WIN'*

'help build an open science community'

*'I want science to be more transparent,
reproducible, accessible, and inclusive'*

*'ensuring research is
not just shared but is
truly accessible and
actionable'*

Open Win Ambassadors events!

- GIT/GITHUB/GITLAB course and practical

Next reboot camp

- Open science Hack-day

Soon



Thank you!

If you have any questions
open@win.ox.ac.uk

Special thanks

- Cassandra Gould Van Praag
- Dr. Hanna Willis
- Open Win Contributors